

Why this model?

Not the other 47 in the family.



The usual shortcut

Pick the biggest, most powerful or most complex model and call it the worst case.

That is often the wrong question.

One model is not worst-case for every test

A configuration can be the most demanding for one requirement and irrelevant for another.

Temperature / heating

Often driven by:

- higher power input
- component loading
- restricted ventilation
- demanding operating mode

Model A may be worst case

Insulation / construction

Often driven by:

- smaller creepage / clearance
- different insulation system
- thinner walls or barriers
- a lower-rated variant

Model B may be worst case

Higher power is not automatically the worst case. A lower value of a parameter can be more critical.

Map the differences first

Worst-case selection starts with what actually varies — not with a favourite model.

01 Components

Which parts, suppliers or constructions change across the family?

02 Ratings

Power, voltage, current, capacity, speed, load — what moves?

03 Construction & dimensions

Size, materials, barriers, ventilation paths, mounting.

04 Insulation

Insulation system, creepage, clearance, working voltage.

05 Protective devices

Fuses, thermostats, software limits, cut-outs.

06 Controls & modes

Control functions and operating modes that change the stress.

Not every difference matters

Keep only the differences that can influence the test you are evaluating.

Example — temperature-related tests

- **Power input**
Higher dissipation, hotter surfaces
- **Component loading**
Stressed parts, not just the nameplate
- **Ventilation**
Restricted airflow can beat higher power
- **Construction**
Materials, layout, heat paths
- **Operating conditions**
Mode, duty cycle, ambient, load

Other safety tests may be driven by completely different parameters. There may not be one worst-case model for the entire product.

Cover the family without gaps

The goal is not simply to test fewer products.

1

Map

List the differences across the product family.

2

Connect

Link each difference to the tests it can influence.

3

Select

Choose the configurations that give sufficient evidence.

4

Justify

If one result covers many models, write why it is representative.

Where certification strategy starts

Worst-case selection starts with the parameter, not the model.